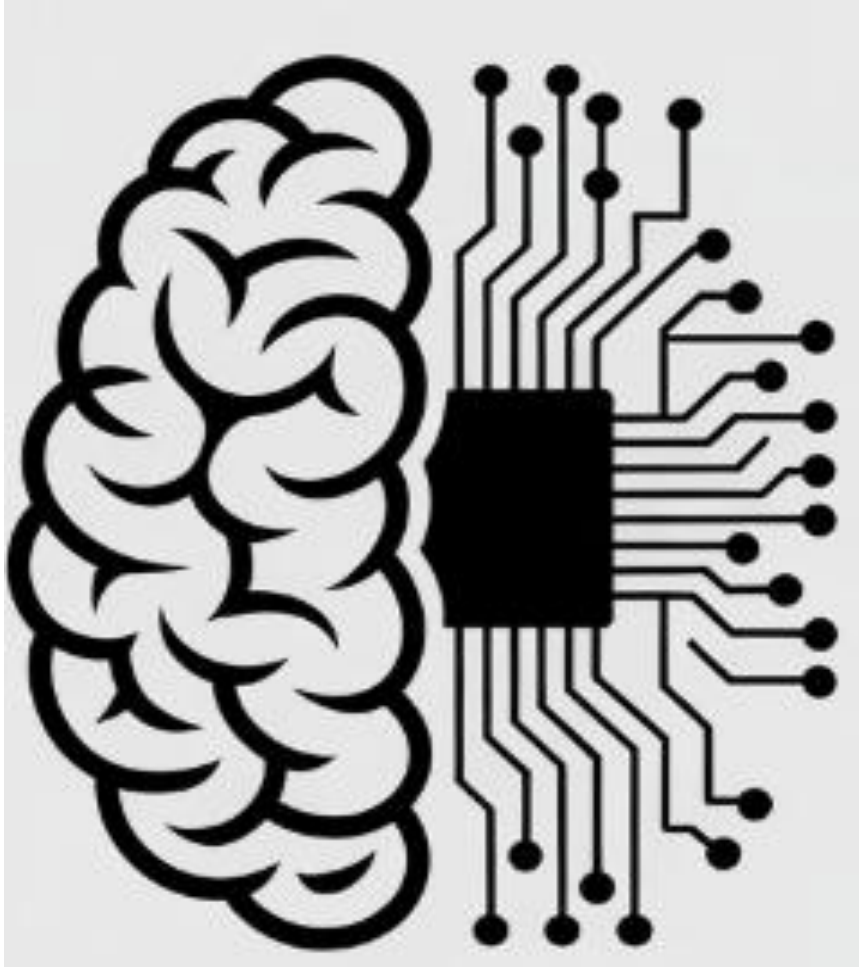




Machine Learning

Dr. Sobia Tariq Javed

ICU Admission Prediction Problem

- An emergency department records **17 clinical variables** for each newly admitted patient
 - (e.g., blood pressure, heart rate, age, laboratory results, etc.).
- A critical decision must be made:
 - **Should the patient be admitted to the Intensive Care Unit (ICU)?**
- Since ICU resources are limited and costly:
 - Patients with **high short-term mortality risk** (e.g., **< 1 month survival**) are prioritized.

ICU Admission Prediction Problem

- Objective: Develop a predictive model to:
 - Identify high-risk patients
 - Discriminate them from low-risk patients
- Key Question:
 - What type of data is being used?

Examples of Data Modalities in AI Applications

- **Image Data:** Object detection, medical imaging analysis
- **Textual Data:** Chat analysis, text generation, word prediction, sentence completion
- **Audio/Speech Data:** Speech recognition, speaker identification, emotion detection
- **Biomedical Signals:** EEG (brain signals), ECG, other physiological signals
- **Medical & Clinical Data:** Lab results, diagnoses, vital signs
- **Numerical/Integer Data:** Counts, scores, encoded categories

Examples of Data Modalities in AI Applications

- Tabular Data (Structured Data):
 - Finance (banking, loans, credit risk)
 - Demographics (age, gender, ethnicity, education)
 - Academic records (GPA, attendance, marks)
 - Online learning analytics (clickstream data from edX, Udemy, LMS logs)

Applications of Machine Learning

- **Spam Filtering**
 - Automatic detection and blocking of unwanted emails
- **Speech & Handwriting Recognition**
 - Converting spoken or handwritten input into digital text
- **Object Detection & Recognition**
 - Identifying and classifying objects in images or videos
- **Weather Forecasting**
 - Predicting atmospheric conditions using historical and real-time data
- **Stock Market Analysis**
 - Trend prediction, risk modeling, and algorithmic trading

Applications of Machine Learning

- **Search Engines**
 - Ranking and retrieving relevant results (e.g., Google)
- **Online Advertising**
 - Personalized ad placement based on user behavior
- **Credit Card Fraud Detection**
 - Identifying suspicious financial transactions
- **Webpage Clustering**
 - Grouping news/articles by topic (e.g., Google News)
- **Social Network Analysis**
 - Community detection, influence modeling, link prediction

Applications of Machine Learning

- **Recommendation Systems**

- Personalized content/product suggestions (e.g., Netflix, Amazon)

- **Autonomous Vehicle Navigation**

- Real-time perception, decision-making, and path planning

Machine Learning definition

- Tom Mitchell (1998) Well-posed Learning Problem:
- A computer program is said to learn from **experience E** with respect to some **task T** and some **performance measure P**, if its performance on T, as measured by P, improves with experience E.
- **Key Components**
 - T (Task): What the system is expected to do
 - E (Experience): The data or interactions used for learning
 - P (Performance Measure): The metric used to evaluate improvement
 - Learning occurs when performance improves with experience.

Learning in Machines

- Similar to **humans learning** from **past experiences**, machines improve performance based on **prior information**.
- A computer does not have experiences in the human sense.
- Instead, it learns from data, which represent **recorded observations** from an application domain.

Learning Objective

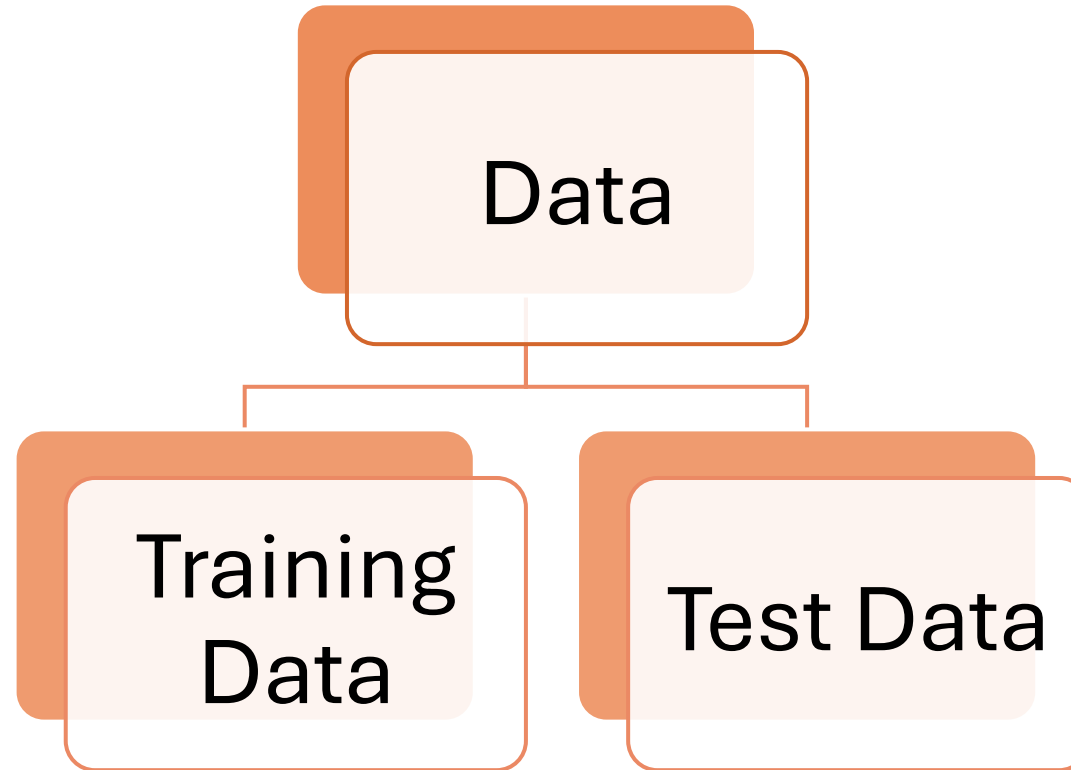
- Learn a **target function** that maps input features to a **discrete class label**.
- **Example class labels:**
 - Approve / Not Approved
 - High Risk / Low Risk
 - Spam/ not spam
- **Type of Learning**
 - This setting is known as **Supervised Learning**
 - When the output is **categorical**, it is specifically called **Classification**

Data and Goal

- **Data:** A set of data records (also called examples, instances or cases) described by
 - **k attributes:** $A_1, A_2, \dots, A_k$.
 - **Class:** Each example is labelled with a pre-defined class.
- **Goal:** To learn a classification model from the data that can be used to predict the classes of new (future, or test) cases/instances.

Data

**teach the
model.**



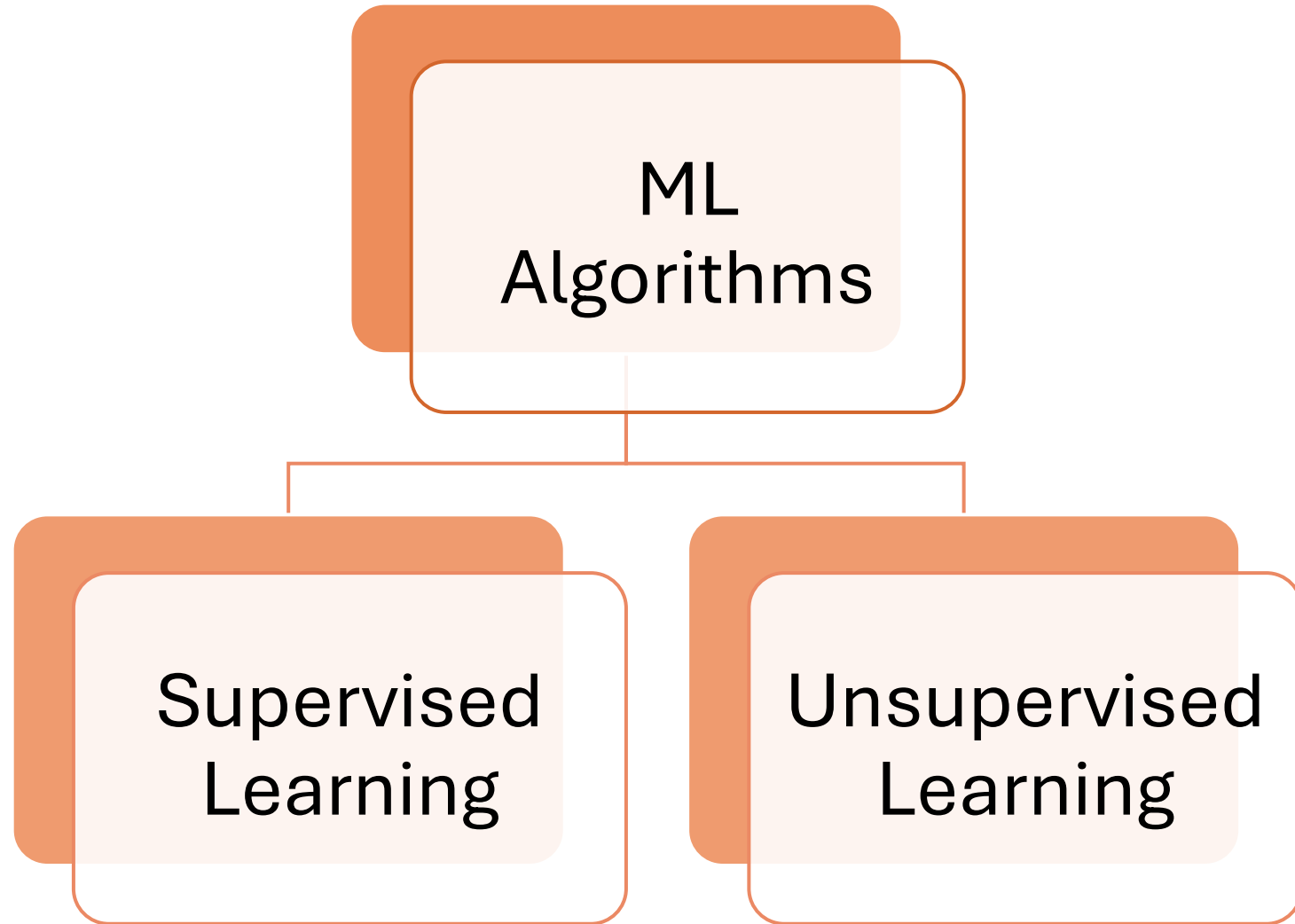
**evaluate the model's
performance after
training**

ICU Admission Prediction Problem

Patient_ID	Age	Systolic_BP	Heart_Rate	Resp_Rate	Oxygen_Sat (%)	WBC_Count ($\times 10^9/L$)	Creatinine (mg/dL)	Diabetes	Predicted_Risk	ICU_Admission
P001	72	85	118	30	88	15.2	2.1	Yes	High	Yes
P002	45	130	82	18	97	7.8	0.9	No	Low	No
P003	63	95	110	26	91	13.5	1.8	Yes	High	Yes
P004	29	120	76	16	99	6.5	0.8	No	Low	No
P005	58	100	102	24	93	11.9	1.5	Yes	High	Yes



Categories of Machine Learning Algorithms



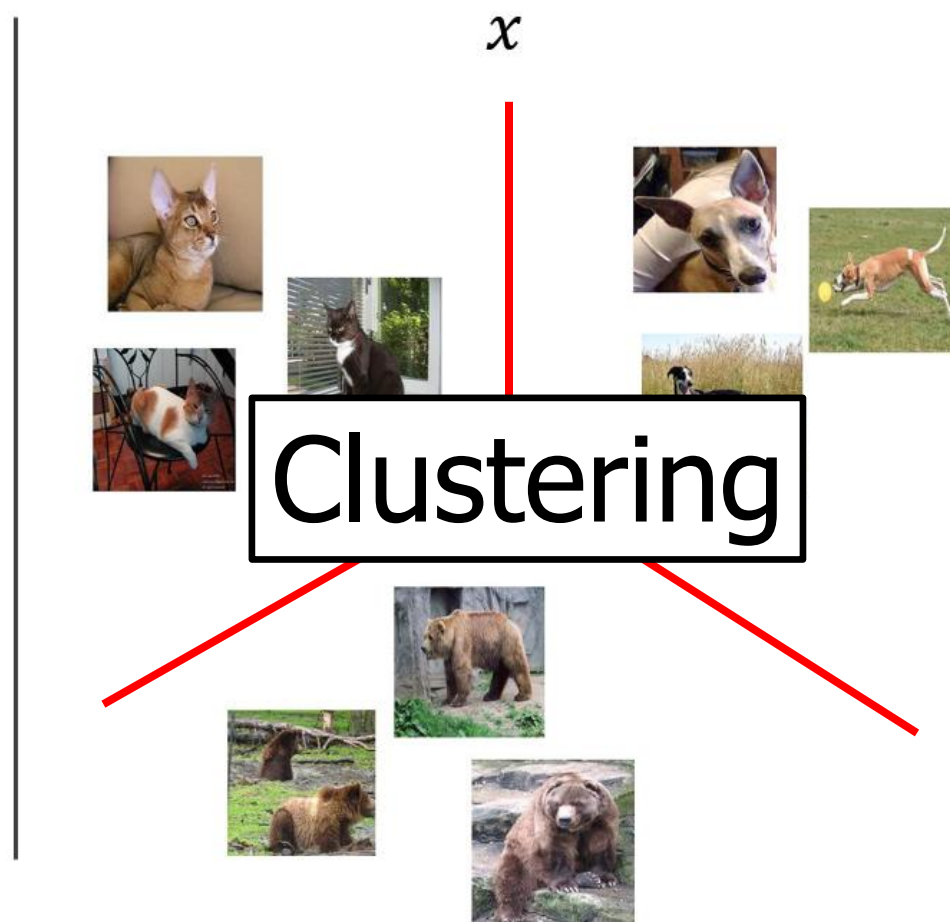
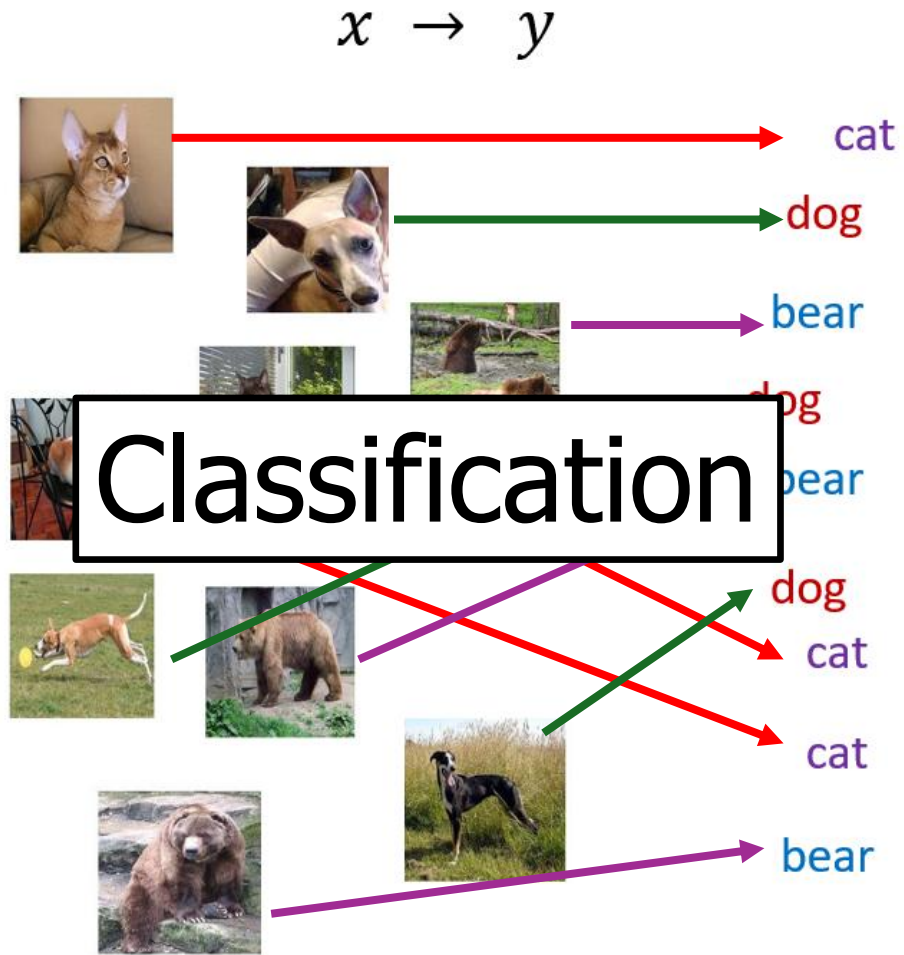
Supervised Learning

- Learning from labeled data
- **Input-output** pairs are provided during training
- **Objective:** learn a mapping function from features to target labels
- **Common tasks:**
 - Classification
 - Regression

Unsupervised Learning

- Learning from unlabeled data
- No predefined target variable
- **Objective:** discover hidden structure or patterns in data
- **Common tasks:**
 - Clustering
 - Dimensionality Reduction
 - Association Rule Mining

Supervised Learning vs Unsupervised Learning



Supervised Learning vs Unsupervised Learning

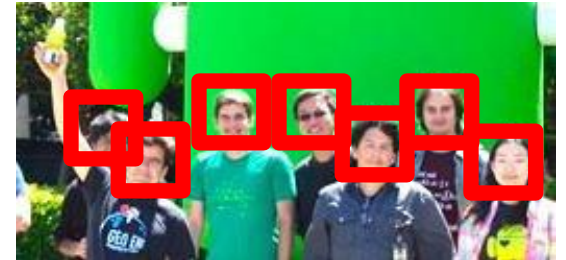


Classification

cat

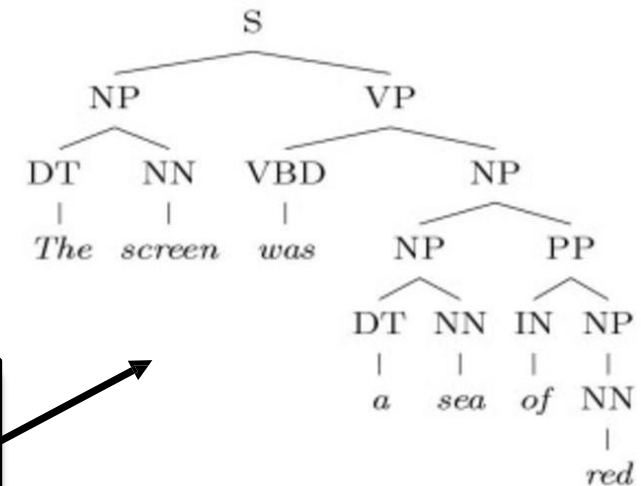


Face Detection



The screen was
a sea of red

Language Parsing



Structured Prediction

Supervised Learning

Given an input $\mathbf{x}$, the goal is to predict a corresponding label $\mathbf{y}$.

Forms of Input $\mathbf{x}$

- Input data may take different representations:
 - **Feature Vectors**
 - $\mathbf{x} \in \mathbb{R}^n$ (tabular/numerical data)
 - **Sequences**
 - Natural language text, time-series signals
 - **Matrices**
 - Images (pixel grids)
 - **Graphs**
 - Nodes and edges, possibly with associated features

Mathematical Formulation

- Supervised learning is framed as an optimization problem:

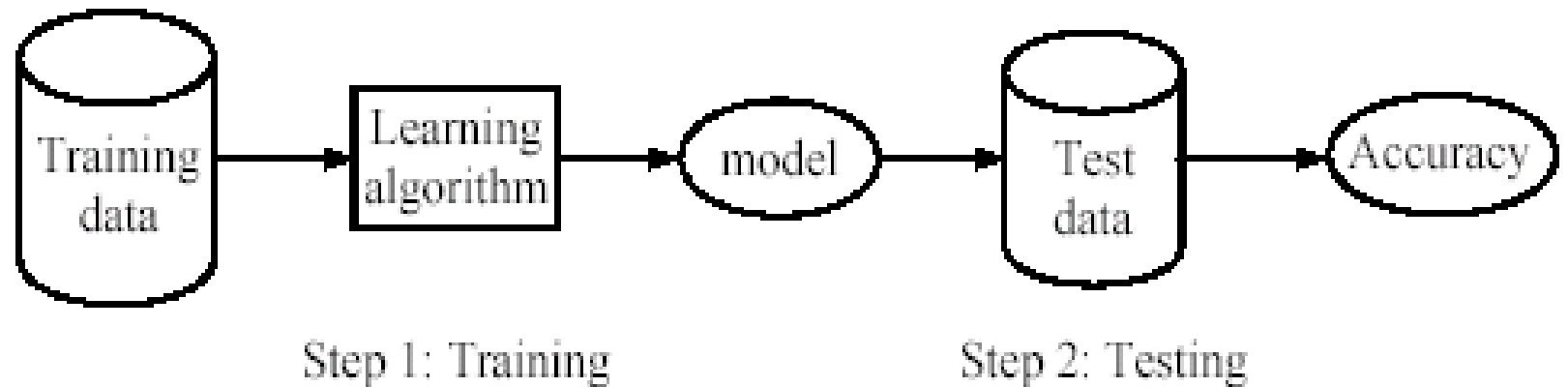
- Learn a function

$$f: x \rightarrow y$$

- Define an objective (loss) function that quantifies the error between predicted and true labels
- Optimize model parameters to minimize the loss over the training data.

Supervised learning process: two steps

- **Learning (training):** Learn a model using the training data
- **Testing:** Test the model using unseen test data to assess the model accuracy



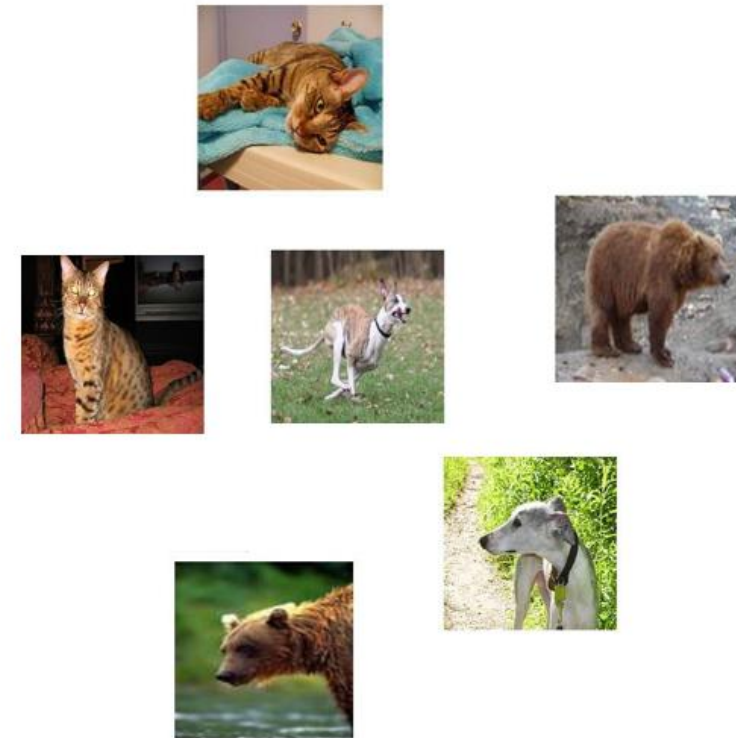
$$Accuracy = \frac{\text{Number of correct classifications}}{\text{Total number of test cases}}$$

Supervised Learning - Classification

Training Data



Test Data



Fundamental assumption of learning

- **Key Assumption in Supervised Learning**

- The training data distribution is assumed to be identical to the test data distribution (including future unseen samples).

- **Reality in Practice**

- This assumption is often partially violated.
- Significant distribution mismatch (data shift) can lead to:
 - Poor generalization
 - Reduced classification accuracy

- **Implication**

- To achieve strong performance on test data, the training set must be sufficiently representative of the real-world data the model will encounter.

Supervised Learning - Classification

Training Data



cat



dog



cat

•
•
•



bear

Test Data



•
•
•



Supervised Learning - Classification

Training Data			
inputs			
$x_1 = [x_{11} \ x_{12} \ x_{13} \ x_{14}]$	targets / labels / ground truth $y_1 = 1$	predictions $\hat{y}_1 = 1$	
$x_2 = [x_{21} \ x_{22} \ x_{23} \ x_{24}]$	$y_2 = 2$	$\hat{y}_2 = 2$	
$x_3 = [x_{31} \ x_{32} \ x_{33} \ x_{34}]$	$y_3 = 1$	$\hat{y}_3 = 2$	
.			
.			
.			
$x_n = [x_{n1} \ x_{n2} \ x_{n3} \ x_{n4}]$	$y_n = 3$	$\hat{y}_n = 1$	

- We need to find a function that maps x and y for any of them.

$$\hat{y}_i = f(x_i; \theta)$$

Learnable
parameters

- How do we "learn" the parameters of this function?
- We choose ones that makes the following quantity small:

$$\sum_{i=1}^n \mathcal{L}(\hat{y}_i, y_i)$$

Loss function

Supervised Learning

**Predict continuous
valued output (price)**

**Supervised
Learning**

**Predict discrete
valued output (0 or 1)**

Regression

Classification

